

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Overview

Imagine you are a user interacting with a chatbot service (e.g., Gemini, ChatGPT, or DeepSeek) for help with programming tasks. On each page, you will see:

- **A Question:** This could be a vague user query or a code snippet requiring clarification or completion. Questions containing code will be displayed in code blocks for clarity.
- **Two Responses:** labeled Answer A and Answer B

Your Task

Please evaluate Answer A and Answer B based on the following criteria:

- **Precision and Focus:** How directly the response targets the question's core issue, using clear guidance, examples, or code.
- **Contextual Fit:** How well the response matches the user's programming goal (e.g., debugging, feature implementation).
- **Answer Faithfulness:** How effectively the response provides actionable steps or code to achieve the user's desired outcome.
- **Correctness:** Correctness in LLM responses refers to the accuracy and factual validity of the information, code, or guidance provided, ensuring it is free from errors and aligns with the user's query and intended outcome.

How to Rate

For each question, you will provide two types of ratings:

1. **Preference Rating (1–5 scale):**
 - **"1"**: Answer A is better, based on the criteria above.
 - **"5"**: Answer B is better, based on the criteria above.
 - **"3"**: Both answers are equally good, based on the criteria above.

Note:

- **Incomplete Responses:** Some responses may appear cut off due to a 256-word limit imposed on the chatbots. Please grade only the content provided, ignoring any abrupt endings or non-English text (e.g., stray characters).
- **Code Formatting:** Questions and responses containing code will be presented in code blocks or artifact tags for clarity. Ensure you review these carefully for correctness and relevance.
- **Consistency:** Try to apply the same standards across all evaluations to ensure fair comparisons.

Question 1

I'm stuck on a part of my code.

Answer A:

“Extra context:

User Question: *I'm stuck on a part of my code.*

AI clarification question: *Could you tell me a little bit about what part of your code you're stuck on, and what the code should be doing? Specifically, what's causing you trouble? (e.g., syntax errors, logic errors, etc.)*

User follow up response: *My JavaScript code for a calculator app isn't updating the display after calculations.”*

AI Final answer:

Okay, let's tackle this. You're stuck with JavaScript code that's not updating the display after calculations. This is a very common problem, and it often stems from how you're managing the display.

The Problem:

Your code is likely updating the display (usually using `console.log()` or a web page element) *after* a calculation is performed. However, the calculation itself isn't actually *changing* the display's content. The display is just getting updated *after* the previous calculation.

What the Code *Should* Be Doing (The Logic):

The core idea is to:

1. Perform the calculation: Inside the `calculate` function, you need to perform the math.
2. Update the display: After the calculation, you need to update the element that represents the display (e.g., `document.getElementById('result')` or a specific HTML element).

Here's a basic example (Illustrative - Adjust to your specific needs):

```
JavaScript
function calculate() {

    let num1 = 10;

    let num2 = 5;

    let sum = num1 + num2;

    // Update the display element here, e.g.:
```

```
document.getElementById('result').textContent = sum;

}
```

Answer B:

Please provide me with the code! I need to see it to help you. 😊

I'm here to help you troubleshoot and understand what's going on. Just paste the code here, and tell me:

- What language is it in? (e.g., Python, JavaScript, C++, Java, etc.)
- What are you trying to do? (Describe the overall goal of the code.)
- What's the specific problem you're encountering? (What's not working as expected? Are you getting errors? Is it running but not producing the desired result?)
- What have you tried so far? (Share any code you've already written, even if it doesn't seem to be helping.)

The more information you give me, the better I can assist you. Software doesn't work, so please provide the code itself!

Let's get started!

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 2

Contextual Fit (1-5): 1

Answer Faithfulness (1-5): 1

Correctness (1-5): 3

Question 2

Can someone explain how to optimize my code?

Answer A:

Okay, let's talk about optimizing your code! It's a broad topic, and the best approach depends *heavily* on what kind of code you're working with and what you're trying to optimize for. Here's a breakdown of strategies:

1. Understanding the Problem – Profiling & Analysis

- **Don't Optimize Blindly:** The first step is *understanding* where your code is spending time. Just slapping on optimizations without knowing what's slow is a waste of time.
- **Profiling:** This is *essential*. Tools like:
 - **Python:** `cProfile`, `line_profiler`
 - **JavaScript:** Browser developer tools (Performance tab), Node.js's `inspector`
 - **Java:** Java VisualVM, YourKit Java Profiler
 - **C++:** Valgrind, gprof
- **Identify Bottlenecks:** Profiling reveals which functions, lines of code, or even specific operations are consuming the most time. Focus on those first.

2. Algorithmic & Data Structure Optimization

- Choose the right data structures and algorithms for your problem.

Answer B:

“Extra context:

User Question: *Can someone explain how to optimize my code?*

AI clarification question: *What language are you using for your code, and what specific part of it are you finding slow or inefficient? (e.g., loops, data processing, etc.)*

User follow up response: *I'm using C++, and my loop for processing a large dataset is too slow.”*

AI Final answer:

Okay, great! Let's tackle that. A slow loop in C++ is a very common performance bottleneck. Here are some general strategies:

1. **Understand Your Loop's Purpose:**
 - What is the loop supposed to *do*?
 - What's the data structure being iterated over? (e.g., vector, array, list, etc.)
2. **Avoid Unnecessary Copies:**
 - Use references or pointers instead of making copies of data.
3. **Minimize Operations Inside the Loop:**
 - Try to do as much work as possible *before* the loop.
4. **Consider Algorithmic Improvements:**

- Sometimes a different approach or algorithm can dramatically improve performance.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 3

Contextual Fit (1-5): 4

Answer Faithfulness (1-5): 4

Correctness (1-5): 3

Question 3

```
JavaScript
function sendRequest(url) {
    const xhr = new XMLHttpRequest();
    xhr.open('GET', url);
    // TODO: Add onload handler
}
```

Answer A:

“Extra context:

User Question:

```
JavaScript
function sendRequest(url) {...}
```

AI clarification question: *What is the intended purpose of the function sendRequest within the context of web development? Specifically, what action or data are you hoping to send to the URL specified in the function?*

User follow up response: *I want to fetch JSON data from the URL and log it to the console.”*

AI Final answer:

That's a very reasonable and common use case for a `sendRequest` function! The function is designed to make an HTTP GET request to a specified URL and then handle the response from the server, typically to retrieve JSON data.

- `xhr.open('GET', url)`: Initializes the request.
- `xhr.onload`: This event is triggered after the request completes successfully. Inside `onload`, you'll access the response, which will typically be a JSON object you can parse and log.

Answer B:

JavaScript

```
function sendRequest(url) {  
  const xhr = new XMLHttpRequest();  
  xhr.open('GET', url);  
  xhr.onload = function() {  
    console.log('Request successful!');  
    // You can perform actions here after the request is successful.  
    // For example, you could update a UI element, send a notification, etc.  
  };  
  xhr.send();  
}
```

Explanation:

1. `const xhr = new XMLHttpRequest();`:: Creates a new XMLHttpRequest object.
2. `xhr.open('GET', url);`:: Sets up the request.
3. `xhr.onload = function() { ... };`:: Attaches a callback for when the request completes successfully.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 4

Contextual Fit (1-5): 4

Answer Faithfulness (1-5): 3

Correctness (1-5): 3

Question 4

Python

```
import numpy as np
def compute_mean(data):
    return np.mean(data)
# Missing data check
print(compute_mean())
```

Answer A:

Okay, let's clarify what you're trying to achieve with this code snippet. The code appears to be a simple function designed to calculate the mean of a list of numbers. However, it's missing handling for empty lists.

If you try to compute the mean of an empty list, `np.mean()` will raise a `RuntimeWarning: Mean of empty slice`. Here's how you can handle this:

python

Python

```
import numpy as np

def compute_mean(data):
    if not data: # Check if the list is empty
        return None
    return np.mean(data)
```

Answer B:

"Extra context:

User Question:

Python

```
import numpy as np

def compute_mean(data):
    return np.mean(data)

# Missing data check
print(compute_mean())
```

AI clarification question: Are you expecting the function to handle empty lists gracefully, or should it raise an error if the input is empty?

User follow up response: *I want the function to print "empty list" and return None if the list is empty instead of raising an error."*

AI Final answer:

Great! To handle empty lists gracefully, you can check if the input list is empty before computing the mean. Here's the updated function:

```
Python
import numpy as np

def compute_mean(data):
    """
    Computes the mean of a list of numbers.

    Args:
        data: A list of numbers.

    Returns:
        The mean of the numbers in the list, or None if the list is empty.
    """
    if not data: # Check if the list is empty
        print("List is empty.")
        return None
    return np.mean(data)
```

This way, if you pass an empty list, the function returns `None` instead of raising a warning or error.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 1

Contextual Fit (1-5): 3

Answer Faithfulness (1-5): 3

Correctness (1-5): 3

Question 5

I need guidance on a library.

Answer A:

Okay, let's tackle both of those! It's great you're tackling these separately – they're distinct and important.

1. Library - Pandas for Data Analysis

First, let's address the Pandas library. You're right to be struggling with it – it can be a bit daunting at first. Here's a breakdown of how to approach it, and some guidance:

- What is Pandas? Pandas is a Python library for data analysis and manipulation.
- Why is it useful?
 - Data Cleaning: Handling missing values, correcting errors, transforming data.
 - Data Exploration: Examining data distributions, identifying trends, finding outliers.
 - Data Transformation: Creating new columns, aggregating data, performing calculations.
 - Data Analysis: Statistical analysis, visualizations.
- Getting Started:
 - Installation: `pip install pandas`
 - Importing: `import pandas as pd`
 - Reading a CSV: `df = pd.read_csv('yourfile.csv')`
 - Inspecting data: `print(df.head()), print(df.info())`

Answer B:

"Extra context:

User Question: *I need guidance on a library.*

AI clarification question: *Could you clarify whether you mean a software/library package (e.g., Pandas in Python) or a physical/public library?*

User follow up response: *I'm struggling with using the numpy library in Python for data analysis and complex mathematical operations."*

AI Final answer:

NumPy is a fundamental Python library widely used for numerical computing and data analysis. It provides powerful n-dimensional array objects, which are efficient for handling large datasets and performing complex mathematical operations. NumPy arrays support vectorized operations, enabling fast computations without explicit loops.

To get started, you can install NumPy via pip:

Shell

```
pip install numpy
```

Then, import it in your Python script:

Python

```
import numpy as np
```

Here are some key features and tips:

- **Arrays and Matrices:** NumPy's core is the `ndarray` object, which can represent vectors, matrices, or higher-dimensional data.
- **Mathematical Functions:** It includes a wide range of mathematical functions such as linear algebra, Fourier transforms, and random number generation.
- **Broadcasting:** NumPy supports broadcasting, allowing operations on arrays of different shapes.
- **Integration:** NumPy integrates well with other libraries like Pandas, Matplotlib, and machine learning frameworks.

For practical learning, tutorials like the NumPy beginner's guide and interactive exercises on platforms like W3Schools are very helpful. Additionally, video tutorials such as Python NumPy Tutorial for Beginners provide step-by-step introductions.

If you have specific tasks or code you're working on, feel free to share them for more targeted assistance.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 5

Contextual Fit (1-5): 5

Answer Faithfulness (1-5): 3

Correctness (1-5): 3